

Computational Homework Assignments

ASEN 6062: Celestial Mechanics

Spring 2026

Please implement and solve the following computational homework problems using the language / tool of your choice. Details of your coding need not be shown, but can be requested and must be provided on demand. For each problem, please provide the specific information and plots requested. For numerical integrations, you are free to use whatever toolbox you need. Note that MATLAB is available to all students and can be used to implement all of these problems.

Computational HW #1: Due March 13, 2026.

1. Write a simulation program to compute the dynamical motion in the 3-body problem. To do this use the equations of motion expressed in the Jacobi Coordinates introduced in class, and use a dimensionless system defined so that the masses sum to unity ($m_1 + m_2 + m_3 = 1$) and with the gravitational constant taken as unity ($\mathcal{G} = 1$). Your program must be able to take in system parameters (i.e., masses) and initial conditions (i.e., relative positions and relative velocities) and produce a time series of positions and velocities for the dynamical evolution of the system over an arbitrary time. Note that for the given units if you choose your relative positions to be of order unity, then the velocities for bounded motion should also be of order unity.
 - (a) For a set of unequal masses and “arbitrary” initial relative positions and velocities, show that your simulation conserves energy E and angular momentum H at a level consistent with the numerical precision of your approach. To show this, plot the time history of these quantities and show their relative variation from their initial values. You will generally see some small variation due to numerical integration errors – to verify this repeat the same calculation but at a higher numerical tolerance and verify that the small variations consistently decrease in magnitude.
 - (b) For the initial masses $1/2$, $1/3$, $1/6$ place the system into an initial relative equilibrium Lagrange solution, i.e., place them at the vertices of an equilateral triangle with unit distance between them and give them all a common rotation rate about the center of mass of unity. The time history of this system should remain consistent for a while, but as this is an unstable set of masses it will eventually diverge. Repeat, giving a small shift in the initial location of one of the bodies (on the order of 0.001). Now how rapidly does the system depart from its relative equilibrium? Show and compare the resultant trajectories.
 - (c) Repeat the above for initial masses that satisfy the Routh Stability Criterion $m_1 m_2 + m_2 m_3 + m_3 m_1 < 1/27$. Verify that the system now stays close to its initial configuration when given a small shift in its initial conditions. How far can you shift the initial relative positions before it changes significantly from its triangular configuration?

2. Make a copy of the previous program and modify it to simulate motion in the 5-body problem. Again use the relevant Jacobi Coordinate form of the equations of motion, choose the total mass such that $m_1 + m_2 + m_3 + m_4 + m_5 = 1$, etc.

Note, if you wish you can also use the on-line tool REBOUND for this simulation: <https://rebound.hanno-rein.de>

If you use REBOUND, please note this explicitly and also show a verification between your previous 3BP simulation and a REBOUND simulation.

- (a) Verify that your new system also conserves energy and angular momentum, similar to before.
- (b) For an equal mass system, start it in an initially circular orbit for the N -gon central configuration from the HW, making the distance from each body to the barycenter unity. This system should be unstable – comment on how fast it departs from this configuration.
- (c) Repeat the above, but now place the initial system on a mutually eccentric orbit with eccentricity $e = 0.1$. Note, it will be easiest to start such a system at periapsis or apoapsis just by giving the configuration a non-circular spin rate. Comment on how this system evolves as compared to the previous case.
- (d) Experiment with this problem to make a system where a 2-mass and a 3-mass system start in orbit about each other and then flyby each other. Describe how the system evolves – test a number of different initial conditions to sample some of the diversity of outcomes that can occur.